245115: porphyritic microgranite, Kanowna Consols mine

('Red Hill porphyry', Yilgarn Craton granites, Kalgoorlie Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)

Location and sampling

Kurnalpi (SH 51-10), Kanowna (3236)
MGA Zone 51, 366687E 6614391N

Sampled on 22 September 2020

This sample was collected from the 319.45 – 320.37 m depth interval of drillcore RHDD18007, drilled in 2018 by Northern Star Resources Limited. The drillhole is located near the southern edge of the Kanowna Consols open pit, about 9.3 km southeast of Centre Loftys Dam, 5.2 km north of Kanowna Dam, and 4.0 km southwest of Four Mile Hill.

Geological context

The unit sampled is a porphyritic microgranite of the Kalgoorlie Terrane, known informally as the Red Hill porphyry or Red Hill granite (Davis et al., 2010). The Red Hill porphyry is dacitic in composition, about 1500 m long and 250 m wide in plan view, has numerous associated radiating dykes, and hosts gold-mineralized quartz veins (Davis et al., 2010). The Red Hill porphyry is intruded into volcaniclastic conglomerates and sandstones, including a sandstone with a maximum depositional age of c. 2670 Ma (sample LB112, Squire and Cas, 2006). A metadacite collected about 2.2 km to the west yielded an igneous crystallization age of 2708 ± 7 Ma (GSWA 104958, Nelson, 1996). A volcaniclastic sandstone collected about 5.1 km to the west-southwest yielded an igneous crystallization age of 2664 ± 4 Ma (GSWA 245105, Wingate et al., 2021b). A metadacite collected about 2.3 km to the north yielded an igneous crystallization age of 2653 ± 9 Ma (GSWA 222662, Wingate et al., 2021a).

Petrographic description

The sample is a medium-grained, altered, plagioclase-phyric microgranite (Fig. 1), consisting of about 95% groundmass, 3–4% plagioclase phenocrysts, 1% pyrite and trace rutile. The groundmass is a massive, microcrystalline aggregate of interlocking anhedral feldspar and quartz grains up to 0.25 mm in size. Feldspar, most of which is poorly to well-twinned plagioclase, is more abundant than quartz. Much of the groundmass texture is obscured by pervasive phyllic (sericite–carbonate–pyrite) alteration. Plagioclase phenocrysts are subhedral to euhedral, mostly tabular, and up to 2 mm long. Pyrite is subhedral to euhedral, equant, evenly disseminated, and up to 0.4 mm in size. Lath-shaped grains of rutile, up to 0.2 mm long, may be pseudomorphs after original magnetite. The sample is not deformed or obviously metamorphosed.

Zircon morphology

Zircons isolated from this sample are colourless and anhedral to euhedral. Large, anhedral crystals are up to 400 mm long, and typically broken, whereas subhedral to euhedral prisms are up to about 250 µm long, and equant to elongate, with aspect ratios up to 5:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 24–25 June and 8–9 July 2021, using SHRIMP-B. Analyses 1.1 to 19.1 (spot numbers 1–20) were obtained during the first session, together with 14 analyses of the M257 standard, of which 12 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.83% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.30% (1σ). Analyses 20.1 to 2.2 (spot numbers 21–32) were obtained during the second session, together with eight analyses of the M257 standard, of which six analyses indicated an external spot-to-spot (reproducibility) uncertainty of 1.01% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.44% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Thirty-two analyses were obtained from 24 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). Three analyses are >5% discordant. The dates obtained from these three analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 29 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 19 analyses of 14 zircons (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2688 ± 6 Ma (MSWD = 1.5).

Group P comprises 10 analyses of 10 zircons (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 2669–2643 Ma.

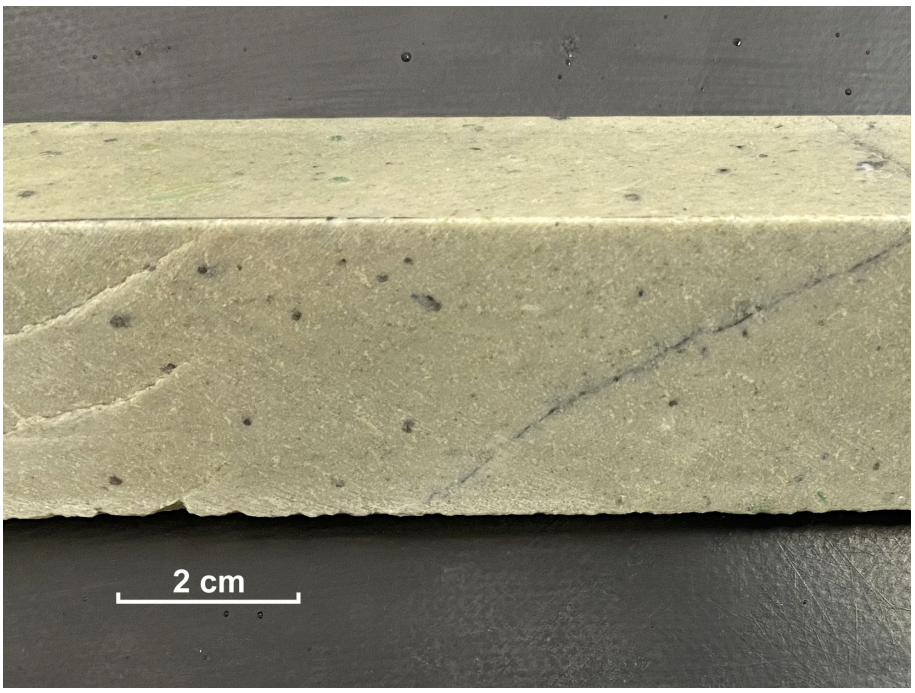

Figure 1. **Drillcore image (wet surface) of sample 245115: porphyritic microgranite, Kanowna Consols mine**

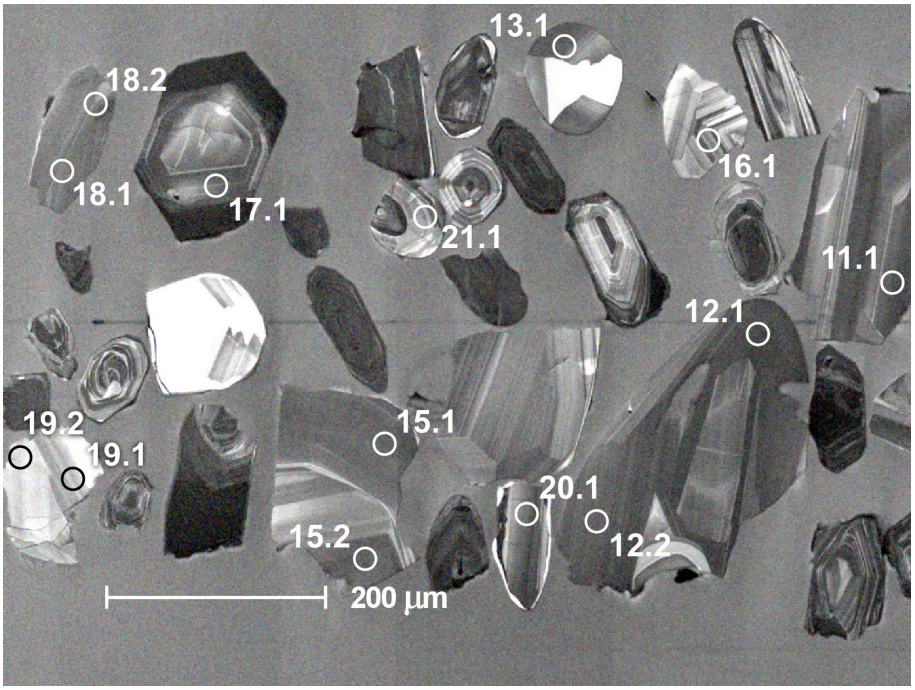

Figure 2. **Cathodoluminescence image of representative zircons from sample 245115: porphyritic microgranite, Kanowna Consols mine. Numbered circles indicate the approximate locations of analysis sites**

Table 1. Ion microprobe analytical results for zircons from sample 245115: porphyritic microgranite, Kanowna Consuls mine

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	$^{238}U/^{206}Pb^*$ date (Ma)	±1σ	$^{207}Pb^*/^{206}Pb^*$ date (Ma)	±1σ	Disc. (%)
I	22	12.2	88	55	0.63	-0.052	2.029	0.030	0.18130	0.00111	2.028	0.029	0.18177	0.00114	2584	32	2669	10	3.2
I	30	24.1	96	64	0.67	0.074	1.996	0.028	0.18258	0.00103	1.998	0.028	0.18192	0.00107	2616	31	2671	10	2.0
I	8	8.1	86	30	0.35	0.180	1.947	0.028	0.18376	0.00128	1.950	0.028	0.18215	0.00140	2668	31	2673	13	0.2
I	14	14.1	72	33	0.46	0.142	1.951	0.030	0.18350	0.00142	1.954	0.030	0.18224	0.00153	2664	34	2673	14	0.3
I	1	1.1	60	17	0.28	0.062	2.003	0.031	0.18280	0.00148	2.004	0.031	0.18225	0.00153	2610	34	2673	14	2.4
I	31	7.2	70	36	0.52	0.025	1.908	0.030	0.18253	0.00133	1.908	0.030	0.18230	0.00135	2716	35	2674	12	-1.6
I	5	5.1	86	54	0.63	0.044	1.961	0.028	0.18351	0.00126	1.962	0.028	0.18311	0.00129	2656	31	2681	12	1.0
I	6	6.1	86	48	0.56	0.042	1.966	0.027	0.18366	0.00123	1.966	0.027	0.18328	0.00126	2650	31	2683	11	1.2
I	12	12.1	181	68	0.37	0.032	1.972	0.023	0.18369	0.00089	1.972	0.023	0.18340	0.00090	2644	26	2684	8	1.5
I	16	15.2	77	45	0.58	-0.024	1.911	0.028	0.18318	0.00132	1.911	0.028	0.18340	0.00134	2713	33	2684	12	-1.1
I	15	15.1	104	60	0.58	0.113	1.975	0.027	0.18470	0.00117	1.978	0.027	0.18370	0.00124	2638	29	2687	11	1.8
I	21	20.1	77	58	0.75	0.120	2.000	0.030	0.18500	0.00120	2.002	0.030	0.18393	0.00128	2612	33	2689	12	2.9
I	27	4.2	32	7	0.23	0.000	1.994	0.038	0.18405	0.00186	1.994	0.038	0.18405	0.00186	2620	42	2690	17	2.6
I	7	7.1	133	91	0.69	-0.043	1.948	0.025	0.18436	0.00102	1.947	0.025	0.18474	0.00104	2672	28	2696	9	0.9
I	32	2.2	96	29	0.30	0.145	2.013	0.029	0.18628	0.00109	2.015	0.029	0.18498	0.00118	2597	31	2698	11	3.7
I	28	22.1	132	89	0.68	0.045	1.990	0.027	0.18544	0.00091	1.990	0.027	0.18504	0.00093	2624	30	2699	8	2.8
I	25	18.2	68	39	0.58	-0.022	1.999	0.030	0.18537	0.00126	1.998	0.030	0.18557	0.00128	2616	33	2703	11	3.2
I	19	18.1	78	43	0.55	-0.023	1.983	0.029	0.18598	0.00131	1.983	0.029	0.18618	0.00133	2633	31	2709	12	2.8
I	2	2.1	115	44	0.39	0.089	1.945	0.025	0.18737	0.00106	1.947	0.025	0.18657	0.00111	2672	28	2712	10	1.5
P	4	4.1	64	20	0.31	0.164	2.002	0.030	0.18043	0.00138	2.005	0.030	0.17896	0.00151	2608	32	2643	14	1.3
P	18	17.1	103	35	0.34	0.087	1.993	0.026	0.18039	0.00110	1.995	0.026	0.17962	0.00115	2620	29	2649	11	1.1
P	17	16.1	48	14	0.30	0.150	1.988	0.033	0.18186	0.00265	1.991	0.033	0.18052	0.00273	2623	36	2658	25	1.3
P	23	21.1	43	23	0.54	0.033	1.958	0.034	0.18088	0.00152	1.959	0.034	0.18058	0.00155	2659	38	2658	14	0.0
P	29	23.1	79	34	0.43	0.020	1.970	0.029	0.18103	0.00118	1.970	0.029	0.18085	0.00120	2646	33	2661	11	0.5
P	20	19.1	46	19	0.41	-0.126	1.985	0.034	0.17974	0.00171	1.982	0.034	0.18086	0.00183	2633	38	2661	17	1.0
P	10	10.1	168	109	0.65	0.034	2.032	0.024	0.18123	0.00090	2.033	0.024	0.18093	0.00091	2579	26	2661	8	3.1
P	9	9.1	85	42	0.50	0.000	2.034	0.029	0.18136	0.00133	2.034	0.029	0.18136	0.00133	2578	31	2665	12	3.3
P	11	11.1	103	86	0.84	0.036	2.042	0.028	0.18201	0.00113	2.043	0.028	0.18169	0.00116	2569	30	2668	11	3.7
P	3	3.1	68	40	0.59	0.000	1.913	0.029	0.18171	0.00135	1.913	0.029	0.18171	0.00135	2711	33	2669	12	-1.6
D	26	19.2	31	9	0.30	0.173	1.819	0.037	0.18280	0.00202	1.822	0.037	0.18126	0.00222	2820	47	2664	20	-5.8
D	24	16.2	34	11	0.32	-0.046	2.085	0.038	0.18169	0.00180	2.084	0.038	0.18210	0.00185	2527	39	2672	17	5.4
D	13	13.1	74	1	0.01	0.133	1.963	0.029	0.20239	0.00150	1.966	0.029	0.20121	0.00159	2651	33	2836	13	6.5

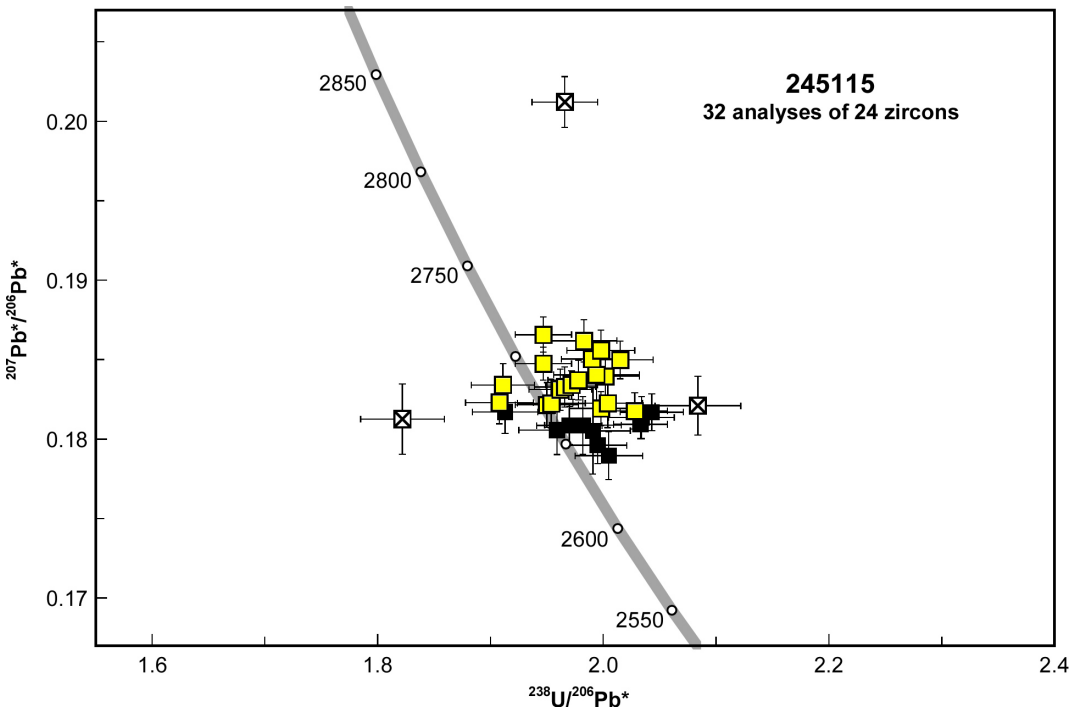

Figure 3. U–Pb analytical data for sample 245115: porphyritic microgranite, Kanowna Consols mine. Yellow squares indicate Group I (magmatic zircons); black squares indicate Group P (radiogenic-Pb loss); crossed squares indicate Group D (discordance >5%)

The date of 2688 ± 6 Ma for the 19 analyses in Group I is interpreted as the igneous crystallization age of the microgranite. The dates of 2669–2643 Ma for the 10 analyses in Group P are interpreted to reflect loss of radiogenic Pb.

References

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Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Smithies, RH 2021, 245115: porphyritic microgranite, Kanowna Consols mine; Geochronology Record 1836: Geological Survey of Western Australia, 4p.

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